

Matheuristics for the Temporal Bin Packing Problem

Fabio Furini and Xueying Shen

Abstract We study an extension of the Bin Packing Problem, where items consume the bin capacity during a time window only. The problem asks for finding the minimum number of bins to pack all the items respecting the bin capacity at any instant of time. Both a polynomial-size formulation and an extensive formulation are studied. Moreover, various heuristic algorithms are developed and compared, including greedy heuristics and a column generation based heuristic. We perform extensive computational experiments on benchmark instances to evaluate the quality of the computed solutions with respect to strong bounds based on the linear programming relaxation of the proposed formulations.

Keywords Temporal bin packing problem • Heuristic algorithms • Column generation

19.1 Introduction

The Bin Packing Problem (BPP) [11] is a classical NP-hard combinatorial optimization problem and it has been widely studied in the literature, we refer the interested readers to [6, 5]. It has applications in different areas, such as filling up containers, placing data on multiple disks and many others. Numerous BPP variants have been studied as the Multi-Dimensional Bin Packing Problem or the On-line Bin Packing Problem. In this paper, we study a natural generalization of the BPP called *Temporal Bin Packing Problem* (TBPP). The input of the problem are a bin capacity W and a set of items $N = \{1, 2, \dots, n\}$, where each item $i \in N$ is associated to an integer weight $w_i \leq W$ consuming the bin capacity during the time window $[s_i, t_i]$. The TBPP asks for finding the minimum number of bins to pack

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all the items so that the capacity of each bin is not exceeded at any point in time. Despite the capacity requirements being defined on the entire time horizon, clearly it is sufficient to satisfy the capacity restrictions only at the starting time of each item. Let us define $S_j := \{i \in N : s_i \leq s_j \text{ and } t_i > s_j\}$ as the set of active items at time s_j , then for each bin, the total capacity of packed items that belong to S_j should be less than or equal to the bin capacity W for any $j \in N$. Moreover, if $S_j \subseteq S_k$, then the associated capacity constraint at time s_j is dominated by that of time s_k . Define $T = \{t \in N : S_t \not\subseteq S_k, \forall k \in N\}$, which represents the index set of all the non-dominated constraints, then we only need to consider the capacity usage at each $t \in T$, and we call t a *time step*.

In Fig. 19.1, we give one example to illustrate a TBPP instance with a bin capacity $W = 4$ and 5 items (weights $w_1 = 2$, $w_2 = 2$, $w_3 = 3$, $w_4 = 2$ and $w_5 = 1$). The starting time and ending time (s_i and t_i) are also shown in the Fig. 19.1. For instance, item 1 is active from time point 1 to 3 and item 2 is active from time point 2 to 14. The set of non-dominated time steps are $t = 2, t = 7$ and $t = 12$, with the following simultaneously active item sets $S_2 = \{1, 2\}$, $S_4 = \{2, 3, 4\}$ and $S_5 = \{2, 5\}$.

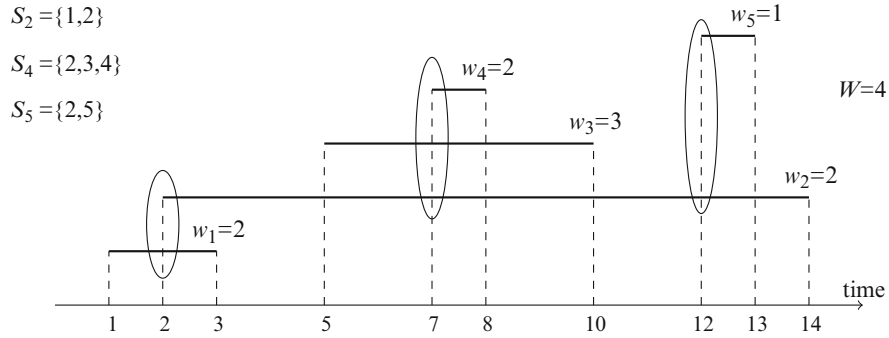


Fig. 19.1 Example of a temporal bin packing problem instance

In fact, the TBPP can be seen as a special case of the multi-dimensional bin packing problem where each time step corresponds to one dimension. Moreover, for each item $i \in N$, it either consumes zero capacity or w_i for each dimension. The concept of “temporal extension” has already been used to extend the classical knapsack problem, which is called Temporal Knapsack Problem (TKP) [1]. TKP has been shown to be strongly NP-hard in [2] and different exact algorithms have been developed, we address the interested reader to [1, 3].

19.1.1 Greedy-Type Algorithms

Inherited from the BPP, we extend two greedy-type algorithms for the TBPP: the first-fit algorithm and the best-fit algorithm.

- First Fit-Algorithm: pack the current item into the first non-empty bin in which it fits. If no such bin exists, pack the current item into an empty bin.
- Best Fit-Algorithm: pack the current item into the non-empty bin with the largest free space in which it fits. By “the largest free space”, we mean the summation of the remaining capacities at all the time steps. If no such bin exists, pack the current item into an empty bin.

These two algorithms are computationally efficient, therefore they can be used to provide the number of bins m for the Mixed Integer Programming (MIP) formulation (see the following sections), or to provide an initial solution for more advanced heuristic algorithms.

The rest of the paper is organized as follows. The compact formulation is given in Sect. 19.2, while an extensive formulation and a heuristic algorithm based on it are presented in Sect. 19.3. The computational results are given in Sect. 19.4, which is followed by the conclusion in Sect. 19.5.

19.2 Compact Formulation

Let $m \leq n$ be an upper bound on the number of bins necessary to fit all items. For each item $i \in N$ and bin $b \in M = \{1, 2, \dots, m\}$, we introduce two sets of binary variables:

$$x_i^b = \begin{cases} 1 & \text{if item } i \text{ is packed in bin } b, \\ 0 & \text{otherwise,} \end{cases} \quad y^b = \begin{cases} 1 & \text{if bin } b \text{ is used,} \\ 0 & \text{otherwise.} \end{cases}$$

Then the polynomial-size MIP formulation reads as follows:

$$\text{Form_A} \quad \min \sum_{b \in M} y^b \quad (19.1)$$

$$\sum_{b \in M} x_i^b \geq 1 \quad i \in N, \quad (19.2)$$

$$\sum_{i \in S_t} w_i x_i^b \leq W y^b \quad b \in M, t \in T, \quad (19.3)$$

$$x_i^b \in \{0, 1\} \quad i \in N, b \in M, \quad (19.4)$$

$$y^b \in \{0, 1\} \quad b \in M. \quad (19.5)$$

Exploiting the link between the TBPP and the BPP, we can compute a valid lower bound for the TBPP from the solution values of $|T|$ BPPs, one for each time step separately. The result is shown in the following property:

Property 19.1. Given an instance of the TBPP, for $t \in T$ we define the $BPP(t)$ as follows:

$$\begin{aligned}
BPP(t) \quad & \min \sum_{b \in M} y^b \\
& \sum_{b \in M} x_i^b \geq 1 & i \in N, \\
& \sum_{i \in S_t} w_i x_i^b \leq W y^b & b \in M, \\
& x_i^b \in \{0, 1\} & i \in N, b \in M, \\
& y^b \in \{0, 1\} & b \in M,
\end{aligned}$$

and let $z^*(BPP(t))$ be the optimal objective function value of $BPP(t)$. Then,

$$\max_{t \in T} z^*(BPP(t))$$

is a valid lower bound of the TBPP.

Proof. For any $t \in T$, the $BPP(t)$ results from relaxing the capacity constraints (19.3) for each $b \in M$ on all time steps except the one at time step t . Therefore, it is a relaxation problem of the TBPP and provides a valid lower bound. The result follows. $\square$

The following example shows that the optimal solution of the TBPP may differ from the optimal solutions of the $BPP(t)$ for each $t \in T$. Given the bin capacity $W = 10$, and 5 items with weights: $w_1 = 9$, $w_2 = 2$, $w_3 = 4$, $w_4 = 6$ and $w_5 = 8$. We consider 2 time steps defined as $S_1 = \{1, 2, 3\}$ and $S_2 = \{2, 3, 4, 5\}$. Then we have $\max_{t=1,2} z^*(BPP(t)) = 2$ (i.e., 2 bins are necessary at any time step), but the optimal solution value of the TBPP is 3.

Besides the lower bound given above, the Linear Programming (LP) relaxation of *Form_A* also provides a valid lower bound. Moreover, the following property shows that the LP relaxation of above formulation *Form_A* can be solved in a closed form.

Property 19.2. Given an instance of the TBPP, the optimal objective function value of the LP relaxation of the formulation *Form_A* can be given in the following closed form:

$$\bar{z}_{lp} = \frac{\max_{t \in T} \sum_{i \in S_t} w_i}{W}, \quad (19.6)$$

with the solution

$$\begin{aligned}\bar{x}_i^b &= \frac{1}{m} \quad i \in N, b \in M, \\ \bar{y}^b &= \frac{\max_{t \in T} \sum_{i \in S_t} w_i}{W \cdot m} \quad b \in M.\end{aligned}$$

Proof. First, we show that the solution is feasible. For any $t \in T, b \in M$,

$$\sum_{i \in S_t} w_i \bar{x}_i^b = \sum_{i \in S_t} w_i \cdot \frac{1}{m} \leq \max_{t \in T} \sum_{i \in S_t} w_i \cdot \frac{1}{m} = \frac{\max_{t \in T} \sum_{i \in S_t} w_i}{W \cdot m} \cdot W = W \bar{y}^b.$$

For any $i \in N$,

$$\sum_{b \in M} \bar{x}_i^b = \sum_{b \in M} \frac{1}{m} = 1.$$

Therefore, the solution $(\bar{x}_i^b, \bar{y}^b)$ is feasible, and the objective function value can be calculated directly as $\bar{z}_{lp}$.

Second, we show that the solution is optimal. For any $t \in T$, we have $BPP(t)$ is a relaxation of $Form_A$, the LP relaxation of $BPP(t)$ is a relaxation of the LP relaxation of $Form_A$. Let $z_{lp}(Form_A)$ and $z_{lp}(BPP(t))$ denote the optimal objective function value of the LP relaxation of $Form_A$ and $BPP(t)$ respectively. Then

$$z_{lp}(Form_A) \geq z_{lp}(BPP(t)) = \frac{\sum_{i \in S_t} w_i}{W}.$$

The second equality is proved in [13]. Take $t^* = \operatorname{argmax}\{t \in T : \max_{t \in T} \sum_{i \in S_t} w_i\}$, the results follows. $\square$

The formulation $Form_A$ is compact in the sense that it has a polynomial number of variables and constraints. An MIP solver can be directly used to solve it, or can be used to obtain heuristic solutions by imposing a given time limit. However, the performance of the formulation highly depends on the upper bound m due to its impact on the number of variables and constraints. Also, the symmetric structure of the model generates a huge amount of equivalent solutions. For large scale problems, MIP solvers are not able to find optimal solutions within reasonable amount of time. Due to these limitations, we propose an extensive formulation which, together with column generation techniques, can be used to develop exact algorithms and heuristic algorithms. Column generation is a powerful tool to deal with LP problems with exponential number of variables. It has been successfully applied to many classes of problems including the cutting stock problem [8, 9], the vehicle routing problem [4], the crew scheduling problem [12] and many others. In the following of the paper, we focus on an extensive formulation of the problem and a heuristic algorithm

based on column generation. We also computationally show that the LP relaxation of the extensive formulation provides tighter bounds, which can be used to prove the optimality of heuristic solutions.

19.3 Extensive Formulation

Let us introduce the extensive formulation with an exponential number of variables associated with all feasible packing patterns, i.e., subsets of items respecting bin capacities at any point in time. Let $\mathcal{S}$ represent the collection of all possible feasible packing patterns:

$$\mathcal{S} = \left\{ S \subseteq N : \sum_{i \in S_t \cap S} w_i \leq W, \forall t \in T \right\}.$$

For each $S \in \mathcal{S}$, we introduce a binary variable z_S :

$$z_S = \begin{cases} 1 & \text{if pattern } S \text{ is selected,} \\ 0 & \text{otherwise.} \end{cases}$$

Then the extensive formulation of the TBPP reads as follows:

$$Form_B(\mathcal{S}) \quad \min \sum_{S \in \mathcal{S}} z_S \quad (19.7)$$

$$\sum_{S \in \mathcal{S}: i \in S} z_S \geq 1 \quad i \in N, \quad (19.8)$$

$$z_S \in \{0, 1\} \quad S \in \mathcal{S}. \quad (19.9)$$

The LP relaxation of $Form_B(\mathcal{S})$ is formulated as ((19.7)-(19.8), (19.10)):

$$z_S \geq 0 \quad S \in \mathcal{S}, \quad (19.10)$$

and it is denoted by $MLP(\mathcal{S})$. The model needs column generation techniques to be efficiently managed. Actually, it has exponentially many variables, which cannot be explicitly generated for large-size instances.

To solve the LP relaxation $MLP(\mathcal{S})$, we start with a restricted problem $MLP(\mathcal{S}')$ with an initial feasible set of variables associated to $\mathcal{S}' \subseteq \mathcal{S}$. The dual problem of $MLP(\mathcal{S}')$ reads as follows:

$$\begin{aligned}
DLP(\mathcal{S}') \quad & \max \sum_{i \in N} \pi_i \\
& \sum_{i \in S} \pi_i \leq 1 \quad S \in \mathcal{S}', \\
& \pi_i \geq 0 \quad i \in N.
\end{aligned}$$

Finding a variable with negative reduced cost to be added to $MLP(\mathcal{S}')$ is equivalent to finding a violated constraint in $DLP(\mathcal{S}')$. Given an optimal dual solution $\pi^* = \{\pi_i^*\}$ to $MLP(\mathcal{S}')$, the separation problem asks to determine a subset $S^* \subseteq \mathcal{S}$ such that $\sum_{i \in S^*} \pi_i^* > 1$. The problem arising during the separation procedure is a 0-1 TKP. This problem can be modelled using a binary variable x_i which takes value 1 if item $i \in N$ is selected in subset S^* and it reads as follows:

$$\max \sum_{i \in N} \pi_i^* x_i \quad (19.11)$$

$$\sum_{i \in S_t} w_i x_i \leq W \quad t \in T, \quad (19.12)$$

$$x_i \in \{0, 1\} \quad i \in N. \quad (19.13)$$

If a feasible pattern S^* with maximum profit greater than 1 is found, the column corresponding to S^* is added to the current $MLP(\mathcal{S}')$. When no feasible pattern with maximum profit greater than 1 is found, $MLP(\mathcal{S})$ is optimally solved, and the rounded-up value of the optimal objective function value to $MLP(\mathcal{S}')$ is a valid lower bound for the TBPP.

In order to find an optimal integer solution of model $Form_B(\mathcal{S})$, a branch-and-price algorithm is necessary, i.e., the column generation procedure must be performed at each node of the branching scheme. The development of such an algorithm goes beyond the scope of this paper, we instead use this formulation to derive a heuristic algorithm and strong dual bounds for hard TBPP instances.

19.3.1 Column Generation Heuristics

In order to obtain good feasible integer solutions, we test a heuristic strategy based on formulation $Form_B(\mathcal{S})$. The procedure consists of applying an MIP solver to model $Form_B(\mathcal{S}')$, by considering all the variables generated to optimally solve its LP relaxation ($MLP(\mathcal{S})$). This strategy takes full advantage of the advanced routines embedded in MIP solvers. The pseudo-code of our heuristic algorithm is given in Fig. 19.2.

begin

Initialize the set $\mathcal{S}'$ with the initial heuristic solution.

1. Solve the $MLP(\mathcal{S}')$ by column generation, and update set $\mathcal{S}'$ by adding the generated columns (cutting patterns);
 2. Solve $Form_B(\mathcal{S}')$ with an MIP solver by considering the subset of variables z_S , $S \in \mathcal{S}'$.
- end.**

Fig. 19.2 Heuristic algorithm based on column generation

The idea of generating a subset of variables containing at least one feasible solution, and then solving the corresponding restricted model with an MIP solver, is an effective method to derive strong upper and lower bounds for the TBPP (see the next section of this manuscript for further details). Moreover, this framework can be applied to different models with an exponential number of variables. We address the interested reader to [7], where similar ideas have been applied to tackle 2-Dimensional BPP.

19.4 Computational Experiments

In this section, we present our computational results for the heuristic algorithms and lower bounds developed above. All the algorithms described in the previous sections are implemented in C++ and run on one core of an Intel Core i7-4790 2.50 GHz 3.60, with 16 GB shared memory, under the Linux Ubuntu 12.4 operating system. All the LP problems and the MIP problems are solved with CPLEX 12.6 [10]. In the following tables, all the computing times are expressed in seconds.

As testbed, we use the instances of the TKP presented in [3], due to the fact that the TBPP instances share the same data structure with those of the TKP, except the absence of item profits. The testbed consists of two types of instances, which are called the I instances and the U instances respectively. Each set consists of one hundred instances which are further divided into ten classes based on the different values of instance generating parameters (see [3] for further details).

We apply the two greedy-type heuristic algorithms as well as the $Form_A$ with a given time limit of 5 min. The detailed results for each instance can be found in Tables 19.1 and 19.2. For each instance, the number of items n and the number of time steps $|T|$ are reported. Moreover, the objective function value and computation time are given in columns *Obj* and *Time*. For comparison, we summarize the results in Table 19.3. In this table, for each class of instances, the average number of items and average number of time steps are reported to illustrate the problem size. We present the average objective function values to give an idea on the number of bins necessary to solve the instances. Clearly the model $Form_A$ benefits from low values of m . Also given are the average computational time and the average gap. The column *Exit Gap* reports the average ratio $\frac{UB-LB}{UB}$ where UB and LB are the primal and the dual bounds computed by CPLEX respectively at time limit (0.00 means instances

Table 19.1 Heuristic algorithms comparison and computational behavior of formulation *Form_A*: Instance U

Inst	n	T	<i>Form_A</i>		Best fit		First fit		Inst	n	T	<i>Form_A</i>		Best fit		First fit	
			Obj	Time	Obj	Time	Obj	Time				Obj	Time	Obj	Time	Obj	Time
U1	1000	459	2	0.02	2	0.01	2	0.00	U51	1000	445	6	0.20	6	0.00	6	0.01
U2	1000	468	2	0.02	2	0.01	2	0.00	U52	1000	467	5	0.18	6	0.01	5	0.00
U3	1000	485	2	0.02	2	0.00	2	0.00	U53	1000	464	6	0.20	6	0.00	6	0.00
U4	1000	484	2	0.02	2	0.01	2	0.01	U54	1000	443	5	0.23	6	0.00	6	0.00
U5	1000	474	2	0.01	2	0.01	2	0.00	U55	1000	456	5	0.17	6	0.00	5	0.00
U6	1000	465	2	0.07	3	0.00	3	0.00	U56	1000	457	6	0.20	6	0.00	6	0.00
U7	1000	487	2	0.02	2	0.01	2	0.01	U57	1000	437	5	0.17	5	0.00	5	0.00
U8	1000	474	2	0.02	2	0.01	2	0.01	U58	1000	435	6	0.18	6	0.00	6	0.00
U9	1000	473	2	0.02	2	0.01	2	0.00	U59	1000	464	6	0.20	6	0.00	6	0.00
U10	1000	468	2	0.02	2	0.01	2	0.00	U60	1000	447	6	0.20	6	0.00	6	0.00
U11	1000	452	3	0.06	3	0.00	3	0.01	U61	1000	433	7	0.24	7	0.00	7	0.00
U12	1000	466	3	0.05	3	0.01	3	0.01	U62	1000	460	6	0.27	7	0.01	6	0.01
U13	1000	467	3	0.06	3	0.01	3	0.01	U63	1000	439	7	0.24	7	0.00	7	0.00
U14	1000	461	3	0.06	3	0.00	3	0.01	U64	1000	453	6	0.24	7	0.01	6	0.00
U15	1000	470	3	0.07	3	0.00	3	0.01	U65	1000	448	7	0.26	7	0.00	7	0.00
U16	1000	465	3	0.06	3	0.01	3	0.01	U66	1000	477	6	0.23	6	0.01	6	0.01
U17	1000	477	3	0.06	3	0.00	3	0.00	U67	1000	444	6	0.32	6	0.00	6	0.00
U18	1000	466	3	0.08	3	0.01	3	0.00	U68	1000	448	6	0.23	6	0.00	6	0.00
U19	1000	467	3	0.06	3	0.01	3	0.01	U69	1000	473	6	0.24	6	0.01	6	0.00
U20	1000	461	3	0.06	3	0.00	3	0.01	U70	1000	439	6	0.21	6	0.00	6	0.00
U21	1000	445	4	0.09	4	0.01	4	0.01	U71	1000	439	7	0.30	8	0.00	7	0.00
U22	1000	478	4	0.10	4	0.00	4	0.00	U72	1000	456	7	0.31	7	0.00	7	0.01
U23	1000	476	4	0.09	4	0.00	4	0.00	U73	1000	444	7	0.28	7	0.00	7	0.00
U24	1000	471	3	0.08	3	0.00	3	0.00	U74	1000	438	7	0.31	7	0.00	7	0.00
U25	1000	454	4	0.09	4	0.01	4	0.00	U75	1000	465	6	277.14	7	0.01	7	0.00
U26	1000	449	4	0.09	4	0.00	4	0.01	U76	1000	439	7	0.28	7	0.00	7	0.00
U27	1000	456	4	0.09	4	0.00	4	0.01	U77	1000	431	7	0.32	8	0.01	7	0.00
U28	1000	462	3	0.07	3	0.00	3	0.00	U78	1000	456	6	0.30	6	0.00	6	0.01
U29	1000	477	3	0.89	4	0.00	4	0.00	U79	1000	453	8	0.33	8	0.00	8	0.00
U30	1000	457	4	0.09	4	0.01	4	0.00	U80	1000	437	7	0.31	7	0.01	7	0.00
U31	1000	446	4	0.09	4	0.00	4	0.00	U81	1000	409	7	7.01	8	0.00	8	0.00
U32	1000	470	5	0.13	5	0.00	5	0.00	U82	1000	451	8	0.39	8	0.00	8	0.01
U33	1000	463	5	0.12	5	0.00	5	0.00	U83	1000	452	7	8.23	8	0.00	8	0.00
U34	1000	449	4	0.10	4	0.00	4	0.00	U84	1000	447	7	0.41	7	0.00	7	0.00
U35	1000	454	4	0.09	4	0.00	4	0.00	U85	1000	441	7	10.37	8	0.01	8	0.00
U36	1000	455	4	0.11	4	0.00	4	0.00	U86	1000	451	7	7.31	8	0.00	8	0.01
U37	1000	444	4	0.10	5	0.01	4	0.00	U87	1000	452	7	0.36	7	0.00	7	0.00
U38	1000	445	4	0.11	4	0.00	4	0.00	U88	1000	452	7	0.44	7	0.00	7	0.00
U39	1000	472	4	0.10	4	0.00	4	0.00	U89	1000	437	8	0.40	8	0.00	8	0.01
U40	1000	442	5	0.13	5	0.00	5	0.00	U90	1000	430	7	7.96	8	0.00	8	0.00
U41	1000	439	5	0.15	5	0.00	5	0.00	U91	1000	441	8	7.60	9	0.00	9	0.00
U42	1000	475	5	0.15	5	0.00	5	0.00	U92	1000	467	8	51.12	9	0.00	9	0.00
U43	1000	476	5	0.15	5	0.00	5	0.00	U93	1000	450	8	0.43	8	0.00	8	0.00
U44	1000	452	4	0.11	4	0.00	4	0.00	U94	1000	449	8	0.55	9	0.00	8	0.00
U45	1000	469	4	2.66	5	0.00	5	0.00	U95	1000	462	8	0.55	8	0.00	8	0.00
U46	1000	456	5	0.15	5	0.00	5	0.00	U96	1000	446	8	7.94	9	0.00	9	0.00
U47	1000	478	5	0.16	5	0.00	5	0.00	U97	1000	475	8	0.49	8	0.01	8	0.00
U48	1000	465	4	2.92	5	0.00	5	0.00	U98	1000	462	8	0.54	8	0.00	8	0.00
U49	1000	436	5	0.14	5	0.00	5	0.00	U99	1000	444	8	0.45	8	0.00	8	0.00
U50	1000	454	5	0.15	5	0.00	5	0.00	U100	1000	438	8	0.44	8	0.01	8	0.00

Table 19.2 Heuristic algorithms comparison and computational behavior of formulation *Form_A*: Instance I

Inst	n	T	<i>Form_A</i>		Best fit		First fit		Inst	n	T	<i>Form_A</i>		Best fit		First fit	
			Obj	Time	Obj	Time	Obj	Time				Obj	Time	Obj	Time	Obj	Time
I1	2697	2688	12	T.L.	12	0.50	12	0.20	I51	4948	768	26	T.L.	26	0.08	26	0.07
I2	2825	2816	12	T.L.	12	0.58	12	0.23	I52	5769	896	26	T.L.	26	0.14	26	0.08
I3	2953	2944	12	T.L.	12	0.60	12	0.26	I53	6721	1024	28	T.L.	28	0.47	28	0.17
I4	3081	3072	12	T.L.	12	0.71	12	0.29	I54	7382	1152	26	T.L.	26	0.84	26	0.30
I5	3209	3200	12	T.L.	12	0.79	12	0.33	I55	8266	1280	26	T.L.	26	1.15	26	0.33
I6	3337	3328	12	T.L.	12	0.92	12	0.36	I56	9002	1408	29	T.L.	29	1.74	29	0.48
I7	3465	3456	11	T.L.	12	0.95	11	0.39	I57	9865	1536	29	T.L.	29	2.19	29	0.69
I8	3593	3584	11	T.L.	11	1.00	11	0.43	I58	10,000	1664	29	T.L.	29	2.91	29	0.71
I9	3721	3712	12	T.L.	12	1.18	12	0.47	I59	10,000	1792	26	T.L.	26	2.55	26	0.76
I10	3849	3840	12	T.L.	12	1.28	12	0.53	I60	10,000	1920	27	T.L.	28	3.13	27	0.77
I11	4480	2688	16	T.L.	16	1.19	16	0.45	I61	2071	768	30	T.L.	30	0.03	30	0.02
I12	4749	2816	15	T.L.	16	1.30	15	0.51	I62	2422	896	28	T.L.	28	0.05	28	0.03
I13	4887	2944	16	T.L.	16	1.49	16	0.56	I63	2763	1024	27	T.L.	27	0.13	27	0.06
I14	5153	3072	16	T.L.	16	1.67	16	0.63	I64	3103	1152	28	T.L.	28	0.22	28	0.09
I15	5298	3200	16	T.L.	16	1.76	16	0.69	I65	3434	1280	27	T.L.	27	0.35	27	0.12
I16	5547	3328	15	T.L.	15	1.83	15	0.75	I66	3774	1408	27	T.L.	27	0.56	27	0.19
I17	5745	3456	17	T.L.	17	2.27	17	0.85	I67	4164	1536	26	T.L.	26	0.70	26	0.26
I18	5929	3584	17	T.L.	17	2.41	17	0.90	I68	4488	1664	27	T.L.	27	0.85	27	0.31
I19	6208	3712	16	T.L.	16	2.58	16	0.98	I69	4786	1792	27	T.L.	27	1.01	27	0.37
I20	6462	3840	16	T.L.	16	2.77	16	1.03	I70	5142	1920	29	T.L.	29	1.36	29	0.45
I21	4972	2688	19	T.L.	19	1.53	19	0.58	I71	2916	768	29	T.L.	29	0.06	29	0.04
I22	5161	2816	22	T.L.	22	2.06	22	0.67	I72	3424	896	28	T.L.	28	0.13	28	0.05
I23	5423	2944	21	T.L.	21	2.24	21	0.74	I73	3832	1024	28	T.L.	28	0.19	28	0.07
I24	5658	3072	21	T.L.	21	2.45	21	0.82	I74	4316	1152	28	T.L.	29	0.41	28	0.12
I25	5875	3200	20	T.L.	20	2.68	20	0.89	I75	4771	1280	27	T.L.	27	0.64	27	0.19
I26	6112	3328	20	T.L.	20	2.77	20	0.98	I76	5403	1408	30	T.L.	30	0.97	30	0.30
I27	6380	3456	21	T.L.	21	2.99	21	1.09	I77	5793	1536	29	T.L.	29	1.11	29	0.37
I28	6582	3584	20	T.L.	20	3.06	20	1.17	I78	6167	1664	27	T.L.	28	1.25	27	0.47
I29	6817	3712	21	T.L.	21	3.48	21	1.27	I79	6800	1792	28	T.L.	28	1.74	28	0.60
I30	7026	3840	21	T.L.	21	3.69	21	1.38	I80	7241	1920	28	T.L.	28	2.00	28	0.69
I31	6322	2688	24	T.L.	24	2.32	24	0.88	I81	5210	768	28	T.L.	28	0.14	28	0.06
I32	6546	2816	23	T.L.	23	2.47	23	0.95	I82	6057	896	26	T.L.	27	0.20	26	0.08
I33	6895	2944	24	T.L.	24	2.97	24	1.14	I83	6901	1024	29	T.L.	29	0.59	29	0.18
I34	7209	3072	25	T.L.	25	3.50	25	1.22	I84	7737	1152	29	T.L.	29	0.85	29	0.22
I35	7484	3200	25	T.L.	25	3.47	25	1.27	I85	8656	1280	28	T.L.	28	1.27	28	0.38
I36	7784	3328	24	T.L.	24	4.13	24	1.51	I86	9370	1408	28	T.L.	28	1.19	28	0.43
I37	8089	3456	24	T.L.	24	4.55	24	1.67	I87	10,000	1536	27	T.L.	27	2.11	27	0.63
I38	8425	3584	23	T.L.	23	4.77	23	1.82	I88	10,000	1664	27	T.L.	27	2.20	27	0.68
I39	8650	3712	24	T.L.	24	5.50	24	1.96	I89	10,000	1792	29	T.L.	29	2.79	29	0.70
I40	8977	3840	23	T.L.	23	5.53	23	2.18	I90	10,000	1920	29	T.L.	29	2.95	29	0.75
I41	2071	768	26	T.L.	26	0.03	26	0.02	I91	3117	768	31	T.L.	30	0.25	31	0.10
I42	2422	896	26	T.L.	26	0.06	26	0.03	I92	3594	896	33	T.L.	33	0.13	33	0.06
I43	2756	1024	27	T.L.	27	0.14	27	0.06	I93	4176	1024	32	T.L.	32	0.27	32	0.10
I44	3104	1152	27	T.L.	27	0.22	27	0.08	I94	4671	1152	32	T.L.	32	0.40	32	0.13
I45	3433	1280	27	T.L.	27	0.36	27	0.13	I95	5209	1280	31	T.L.	31	0.71	31	0.22
I46	3789	1408	28	T.L.	28	0.60	28	0.20	I96	5628	1408	32	T.L.	32	0.99	32	0.33
I47	4154	1536	29	T.L.	29	0.67	29	0.24	I97	6215	1536	31	T.L.	31	1.23	31	0.43
I48	4476	1664	27	T.L.	27	0.85	27	0.31	I98	6730	1664	33	T.L.	33	1.73	33	0.56
I49	4797	1792	28	T.L.	27	0.99	28	0.37	I99	7172	1792	32	T.L.	31	1.96	32	0.67
I50	5129	1920	27	T.L.	27	1.26	27	0.47	I100	7709	1920	31	T.L.	32	2.30	31	0.90

Time limit (T.L.) = 5 min

Table 19.3 Heuristic algorithms comparison and computational behavior of formulation *Form_A*

Group	n	T	Form A			Best fit			First fit		
			Obj	Time	Exit Gap	Obj	Time	Gap	Obj	Time	Gap
U inst.											
I	1000	473	2.00	0.02	0.00	2.10	0.01	5.00	2.10	0.00	5.00
II	1000	465	3.00	0.06	0.00	3.00	0.01	0.00	3.00	0.01	0.00
III	1000	462	3.70	0.17	0.00	3.80	0.00	3.33	3.80	0.00	3.33
IV	1000	454	4.30	0.11	0.00	4.40	0.00	2.50	4.30	0.00	0.00
V	1000	460	4.70	0.67	0.00	4.90	0.00	5.00	4.90	0.00	5.00
VI	1000	451	5.60	0.19	0.00	5.90	0.00	6.00	5.70	0.00	2.00
VII	1000	451	6.30	0.25	0.00	6.50	0.00	3.33	6.30	0.00	0.00
VIII	1000	445	6.90	27.99	0.00	7.20	0.00	4.52	7.00	0.00	1.67
IX	1000	442	7.20	4.29	0.00	7.70	0.00	7.14	7.70	0.00	7.14
X	1000	453	8.00	7.01	0.00	8.40	0.00	5.00	8.30	0.00	3.75
I inst.											
I	3273	3264	11.80	T.L.	6.97	11.90	0.85	0.91	11.80	0.35	0.00
II	5445	3264	16.00	T.L.	28.39	16.10	1.93	0.67	16.00	0.74	0.00
III	6000	3264	20.60	T.L.	–	20.60	2.70	0.00	20.60	0.96	0.00
IV	7638	3264	23.90	T.L.	–	23.90	3.92	0.00	23.90	1.46	0.00
V	3613	1344	27.20	T.L.	–	27.10	0.52	0.00	27.20	0.19	0.37
VI	8195	1344	27.20	T.L.	–	27.30	1.52	0.37	27.20	0.44	0.00
VII	3614	1344	27.60	T.L.	–	27.60	0.53	0.00	27.60	0.19	0.00
VIII	5066	1344	28.20	T.L.	–	28.40	0.85	0.73	28.20	0.29	0.00
IX	8393	1344	28.00	T.L.	–	28.10	1.43	0.38	28.00	0.41	0.00
X	5422	1344	31.80	T.L.	–	31.70	1.00	0.32	31.80	0.35	0.66

Time limit (T.L.) = 5 min

solved to proven optimality and “–” means no dual bound). The column *Gap* reports the average ratio $\frac{z^A - z^*}{z^*}$, where z^* and z^A are the objective function values of the best known solution and that of the solution given by the corresponding algorithm.

We can see that for both sets of instances, the best fit algorithm and the first fit algorithm have the same results in more than 90% of the cases, while the first fit algorithm is computationally more efficient. Hence, we choose the first fit algorithm to provide the initial solutions for the other heuristic algorithms, including providing the initial solution for *Form_A*. We also point out that no I instance can be solved to proven optimality by the compact model *Form_A* within the time limit, whereas all the U instances are optimally solved.

We now test the column generation based heuristic algorithm introduced in Sect. 19.3.1, together we also compare the different lower bounds of the TBPP discussed in this paper, i.e., the LP relaxation of *Form_A*, *Form_B* and the bound based on the BPP in Property 19.1. The initial solution of the algorithm is provided by the first fit algorithm. Since the U instances are easy to solve, we only consider the I instances. Moreover, I instances are computational hard for *Form_B* as well, we only report results for the subset of instances of which we are able to solve its LP relaxation within 1 h via column generation. The results are presented in Table 19.4. For formulation *Form_A*, in column z_{lp} and *Time*, we report the objective function value of the LP relaxation and the computational time of CPLEX. The lower bound based on the BPP is given in column LB_{BPP} . Computing the best LB_{BPP} requires the solution of $|T|$ BPPs and accordingly it may require a very long computational time. We decide instead to solve only 100 BPPs for randomly sampled time steps

Table 19.4 The column generation based heuristic algorithm and lower bounds of the TBPP

Instance	<i>Form_A</i>		LB_{BPP}	<i>Form_B</i>				
	z_{lp}	Time		z_{lp}	Pricing Time	Time	Cols	Obj
I40	18.35	268.31	20	23	321.71	1034.28	789	23
I44	21.53	34.27	26	26	1862.41	2639.48	1631	27
I45	22.18	46.84	24	27	0.48	0.53	81	27
I61	20.69	300.34	26	26	2001.04	3011.65	1924	40
I64	21.43	41.73	26	27	2237.6	2551.92	1091	28
I71	23.41	130.53	28	28	482.14	494.61	514	29
I72	23.07	21.57	26	27	1676.97	1823.78	985	28
I73	22.96	28.75	28	27.5	794.43	831.12	593	28
I75	23.66	42.71	24	27	8.56	13.33	284	27
I76	23.61	44.17	27	29	885.51	948.59	736	30
I77	23.2	73.38	26	29	32.61	38.47	530	29
I78	22.4	76.32	27	27	1035.21	1295.63	566	27
I82	22.7	26.08	26	26	1888.26	2757.68	904	26
I83	24.46	27.98	28	28	480	564.07	896	29
I86	23.2	64.88	27	28	141.08	161.43	742	28
I88	22.62	65.42	26	27	12.17	12.7	135	27
I89	23.79	73.45	29	29	14.88	23.91	717	29
I90	24.53	74.82	28	29	142.55	229.19	1512	29
I93	26.58	49.61	31	31	1163.19	1235.46	752	32
I95	25.14	75.06	31	31	53.06	81.47	417	31
I96	25.55	88.64	32	32	56.23	88.91	477	32
I98	27.58	131.05	30	33	17.17	26.27	378	33
I99	27.06	142.98	29	31	1322.96	1438.21	664	32

in order to achieve a good compromise between computational time and quality of the bounds. Finally, for *Form_B*, column *Obj* and z_{lp} give the objective function value obtained by the heuristic algorithm proposed in Sect. 19.3.1 and that of the LP relaxation. The pricing time to solve the sub problem TKP during column generation and the total computational time of solving the LP relaxation are reported in column *Pricing Time* and *Time* respectively. The time for solving the final MIP of the column generation heuristic is on average very low except in some rare cases in which this phase can take up to 1 min. In our experiments, the TKP sub problems are solved by CPLEX using Formulation (19.11)–(19.13). Table 19.4 also presents the number of columns generated during the entire process.

It is worth mentioning that the computation of the LP relaxation of *Form_A* is surprisingly high while it can be alternatively computed in negligible time using the formula (19.6) presented in Property 19.2. In any case, this bound is very poor compared to other ones. We observe that in all cases, the extensive formulation returns better lower bounds. Also, the lower bounds based on the BPP are strictly better than the ones provided by *Form_A* in all cases except one. Finally, thanks to our column generation algorithm, we can prove the optimality for 14 instances out of 23 instances considered. In Table 19.4, we highlight objective function values of LP relaxation and that obtained by the heuristic algorithm based on *Form_B* for these 14 instances. Many of the I instances still remain open and we hope this paper may stimulate further research on the TBPP.

19.5 Conclusion

In this paper, we study an extension of the bin packing problem called TBPP and present a compact MIP formulation and an extensive MIP formulation. Also, greedy-type heuristic algorithms are developed, while an MIP solver together with the compact formulation is used as a heuristic given time limits. A heuristic algorithm based on the extensive formulation together with the column generation technique is presented. Computational experiments are discussed, showing the strength of the extensive formulation and the potentiality of column generation based algorithms applied to the TBPP.

For future research, a pseudo-polynomial size formulation could be developed. Also, other heuristic algorithms based on column generation can be studied, such as combining diving strategies and variable fixing.

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